1.

Problem 4.1

$$N = \sum \langle S_n \rangle \leftarrow \text{total number of photons}$$

$$\sum \longrightarrow \left(\frac{1}{8}\right)\int_0^\infty \left(4\pi n^2\, dn\right)(2)$$

$\uparrow$ two independent polarizations

$$\Rightarrow \quad N = \sum \langle S_n \rangle \equiv \frac{8\pi}{8}\int_0^\infty n^2\, dn \; \langle S_n \rangle$$

$$= \pi \int_0^\infty n^2\, dn \left[\frac{1}{\exp\left(\frac{\hbar\omega_n}{\tau}\right) - 1}\right]$$

where $\omega_n = \frac{n\pi c}{L}$

Substituting $\rightarrow x = \frac{\pi \hbar c n}{L\tau}$

$$\Rightarrow \quad dn = \left(\frac{L\tau}{\pi\hbar c}\right) dx$$

$$N = \pi \left(\frac{L\tau}{\pi\hbar c}\right)^3 \underbrace{\int_0^\infty \frac{x^2\, dx}{\exp(x) - 1}}_{2.404}$$

$$\Rightarrow \quad N = 2.404\, \pi^{-2}\, V \left(\frac{\tau}{\hbar c}\right)^3 ; \qquad \text{where } V \equiv L^3$$

2.

Problem 4.2

Solar Constant $\equiv 0.136\ J s^{-1}\ cm^{-2}$

(a) $P \equiv 4\pi \left[1.5 \times 10^{13}\right]^2 \times (0.136)$

solar constant

Distance ~~Radius~~ of Earth-Sun

$\simeq 4 \times 10^{26}\ J s^{-1}$

(b) $J_u = \sigma_B T^4$

$\Rightarrow T = \left(J_u / \sigma_B\right)^{1/4}$

$J_u \equiv$ Energy flux density

$$J_u = \frac{4 \times 10^{26}}{4\pi (7 \times 10^8)^2}\ J s^{-1} m^{-2}$$

radius of the sun

$\sigma_B \equiv 5.6 \times 10^{-8}\ W m^{-2} / K^{-4}$

$T \approx 6000 K$

③

Problem 4.5

Power Earth recieves :

Solar Constant $J_s = 0.136 \text{ J s}^{-1} \text{ cm}^{-2}$

Normal - Cross-section of Earth which "recieves" it

πR_E^2 ; $R_E \equiv$ radius of earth

Normal- Cross-section of Earth which "emits" it

$4\pi R_E^2$

$$\Rightarrow \quad P^{\text{recieved}} = J_s \, \pi R_E^2$$

$$P^{\text{emitted}} = J \, 4\pi R_E^2$$

$$\Rightarrow \quad J = \frac{J_s}{4}$$

$$\Rightarrow \quad J = \sigma_B T_E^4 \; ; \; T_E \equiv \text{surface temperature of Earth.}$$

$$T_E = \left(\frac{J_s}{4\sigma_B} \right)^{1/4} \approx 280 \text{ K}$$

Problem 4.7

5. (a) For a single mode:

$$Z_j = \left[1 - \exp\left(-\hbar\omega_j/\tau\right)\right]^{-1}$$

For a combination of independent modes:

$$Z = Z_1 Z_2 \cdots Z_j \cdots$$

$$= \prod_n \left[1 - \exp\left(-\frac{\hbar\omega_n}{\tau}\right)\right]^{-1}$$

(b) $$F = -k_B T \ln Z$$

$$= -k_B T \ln\left(Z_1 Z_2 \cdots Z_j \cdots\right)$$

$$= -k_B T \sum_n \ln Z_n$$

$$= -k_B T \sum_j \ln\left[1 - \exp\left(-\frac{\hbar\omega_n}{\tau}\right)\right]^{-1}$$

$$= k_B T \sum_j \ln\left(1 - \exp\left(-\frac{\hbar\omega_n}{\tau}\right)\right)$$

$$\sum_j \equiv 2 \times \frac{1}{8} \int 4\pi n^2 dn \equiv \pi \int n^2 dn$$

$$\omega_n = \frac{n\pi c}{L}$$

(1) 3.11

$$\varepsilon_n = \frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 n^2 \equiv \varepsilon_1 n^2 \; ; \quad \text{where } \varepsilon_1 = \frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2$$

for a single particle partition function :

$$Z_1 \equiv \sum_n \exp\left(-\varepsilon_1 \frac{n^2}{\tau}\right)$$

$$\cong \int_0^\infty \exp\left(-\varepsilon_1 n^2/\tau\right) dn$$

$$= \sqrt{\frac{\pi\tau}{4\varepsilon_1}} = n_{Q_1} L \; ; \quad n_{Q_1} = \left(\frac{m\tau}{2\pi\hbar^2}\right)^{1/2}$$

For N particles: $Z_N = \frac{(Z_1)^N}{N!}$

$$F = -\tau \log Z_N$$

$$= -\tau N \log Z_1 + \tau \log N!$$

$$= -\tau N \log(n_{Q_1} L) + \tau N \log N - \tau N$$

$$= -\tau N \log \frac{n_{Q_1} L}{N} - \tau N$$

Note: $\frac{\partial n_{Q_1}}{\partial \tau} = \frac{1}{2}\frac{n_{Q_1}}{\tau} \Rightarrow \sigma = -\left.\frac{\partial F}{\partial \tau}\right)_n = N \log \frac{n_{Q_1}}{n} + \frac{N}{2} + N = N \log \frac{n_{Q_1}}{n} + \frac{3N}{2}.$

$$F = \pi (k_B T) \int_0^\infty n^2 \ln\left(1 - \exp\left[-\frac{\hbar n \pi c}{L k_B T}\right]\right)$$

$$k_B T = \tau$$

choose $\quad x = \dfrac{\hbar \pi c}{L \tau} n$

$$\Rightarrow \quad n = \frac{L\tau}{\hbar \pi c} x$$

$$= \pi \tau \left(\frac{L\tau}{\hbar \pi c}\right)^3 \int_0^\infty x^2 \ln(1 - e^{-x})\, dx$$

[Both limits]

$$\int_0^\infty x^2 \ln(1-e^{-x})\, dx = \cancelto{0}{\frac{x^3}{3} \ln(1-e^{-x}) \Big|_0^\infty}$$

$$- \int_0^\infty \frac{x^3}{3} \frac{[e^{-x}]}{(1-e^{-x})}\, dx$$

$$= -\frac{1}{3} \int_0^\infty \frac{x^3}{e^x - 1}\, dx$$

$$= -\frac{1}{3} \frac{\pi^4}{15}$$

$$\Rightarrow \quad F = -\frac{\pi^2 V \tau^4}{45 \hbar^3 c^3}$$